CARSON SLATER

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SUMMARY

Statistical Science Ph.D. candidate and former collegiate baseball player with a passion for sports analytics. Researcher and consultant who helps stakeholders turn abstract questions into data-driven decisions and frameworks, with experience handling large insurance and utility data.

SKILLS

Languages	Python (pandas/polars, scikit-learn, statsmodels, PyTorch), R (tidyverse, Shiny, Quarto, rstan, rjags, torch), SQL
Tools	Git/GitHub, Docker, DuckDB, LaTeX, reproducible pipelines (targets)
Stats & ML	Bayesian methods, linear mixed models, time series, spatiotemporal modeling, survival analysis, mixture models, deep learning (CNNs, diffusion models), tree-based models
Coursework	Statistical Bioinformatics, Deep Learning, Bayesian Methods/Theory, Survival Analysis, Time Series, Categorical Data Analysis, Multivariate Statistics, Computational Statistics, Probability Theory, Machine Learning

EXPERIENCE

Data Analyst August 2026 - Present
Texas Farm Bureau Insurance Companies, *Part Time* *Waco, TX*

- Built and assessed GLMs as pricing models on 100+ GB datasets to improve the auto insurance rating plan, working within regulatory requirements of the Texas Department of Insurance.
- Ingesting and evaluating new vendor data to assess if purchasing for auto insurance rating would be profitable, evaluating over 200 potential new rating variables.
- Working with millions of policy rows to answer ad-hoc business questions for stakeholders.

Research Assistant January 2024 - July 2026
Baylor University – Research funded by Denver Water *Waco, TX*

- Developed unsupervised learning methods for assessing urban water consumption with IoT meters.
- Extracted, transformed, and loaded unstructured data for analyses for Denver Water collaborators.
- Collaborated with domain experts in urban water demand planning to assess water conservation interventions, implementing rigorous hypothesis testing and building Shiny dashboards to visualize individual and aggregate customer trends.
- Engineered features from household-level billing and water usage time series to perform market segmentation for Denver Water's single-family residential customer base, coming up with seven user personas to characterize their population (∼200k homes) to help drive targeted demand management strategies.
- Edited and reviewed material for an undergraduate intro to data science course.

Statistical Consultant August 2024 - July 2026
Baylor University Statistical Consulting Center *Waco, TX*

- Created and assessed a quantitative framework for assessing the success of college athletics coaches for the Baylor Athletic director. Used selemium to web scrap and compile a large dataset for modeling collegiate coach success using both frequentist and Bayesian mixed effects models.
- Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association.
- Delivered quality reports for each client and had regular meetings to discuss the project, the methodology and results.

Academic Coach January - May 2023
Wheaton Academy Center for Lifelong Learning *West Chicago, IL*

- Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.

Mathematics Teaching Assistant August 2022 - May 2023
Wheaton College *Wheaton, IL*

- Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.
- Graded homeworks for the teacher of record for Probability Theory and Machine Learning.

Biostatistics Fellow June - July 2022
Duke/North Carolina State Summer Institute for Biostatistics *Raleigh/Durham, NC*

- Engaged in undergraduate discussions and biostatistical methods, clinical trials, and statistical programming.
- Participated in a data hackathon studying myocardial infarction to uncover risk factors for a particular population.

EDUCATION

Ph.D. Statistical Science, Baylor University 2024 - 2027 [Expected]
Advised by Amanda S. Hering · Concentrations in *Data Science* and *Biostatistics*
Research involves developing unsupervised learning methods for time series with applications in municipal water distribution.

M.S. Statistical Science, Baylor University 2023 - 2024

B.S. Applied Mathematics with Statistics, Wheaton College (IL) 2019 - 2023

B.A. Economics
Edman Presidential Scholarship
Varsity Baseball Athlete (NCAA-DIII)

LEADERSHIP EXPERIENCE

President May 2026 - Present
Baylor University Statistical Science Paper Club *Waco, TX*

- Sourcing and facilitating discussions on frontier papers in statistics, biostatistics, and machine learning for graduate students in the statistical science department.

President March 2024 - May 2025
Baylor University Sports Analytics Club *Waco, TX*

- Led a team of undergraduates in creating a March Madness bracket challenge submission on Kaggle.
- Taught undergraduates basic version control, R programming, statistical and machine learning topics.

PEER-REVIEWED PUBLICATIONS

Brittany J. Carnathan, Dinny Stevens, Swarna Shikha, **Carson Slater**, Nathen Byford, Rodney X. Sturdivant, Kuzy Zarzosa, W. Evan Braswell, and Christie M. Sayes. 2025. "Assessing the Effects of Surface-Stabilized Zero-Valent Iron Nanoparticles on Diverse Bacteria Species Using Complementary Statistical Models". *Journal of Functional Biomaterials* 16 (3). DOI: 10.3390/jfb16030113

MANUSCRIPTS IN PROGRESS

Carson Slater, Ryan Shepler, Samantha Bryan, Tzahi Y. Cath, and Amanda S. Hering. 2026. "Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption". *In Preparation for Submission to Discover Water*.

CONFERENCE PROCEEDINGS & OTHER WORKS

Carson Slater, Ryan Shepler, Samantha Bryan, Tzahi Y. Cath, and Amanda S. Hering. 2026. "Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption". *The 2026 Joint Statistical Meetings*, Thomas M. Menino Convention & Exhibition Center, Boston, MA, August 1-6, 2026. [Poster Presentation]

Jonathan E. Swindell, **Carson Slater**, Samuel Hussey, Charles Baylis, and Robert J. Marks. 2024. "Assessing Interference with Regression Analysis Techniques". *2024 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS)*. DOI: 10.1109/WMCS62019.2024.10619025.

Amanda S. Hering, Douglas Nychka, Luke Durell, **Carson Slater**, Michael Poor, Greg Hamerly, and Tzahi Y. Cath. 2024. "Mo(Wa)^{2}TER Introduction to Data Science Course Material". *Harvard Dataverse*. DOI: 10.7910/DVN/BZJW2E.

Carson Slater and Ying Li. 2022. "Forecasting Fast-Moving Consumer Goods with Generalized Additive Models". *2022 University of Illinois Chicago Undergraduate Mathematics Symposium*, University of Illinois - Chicago, Chicago, IL, USA, November 12, 2022. [Poster Presentation]